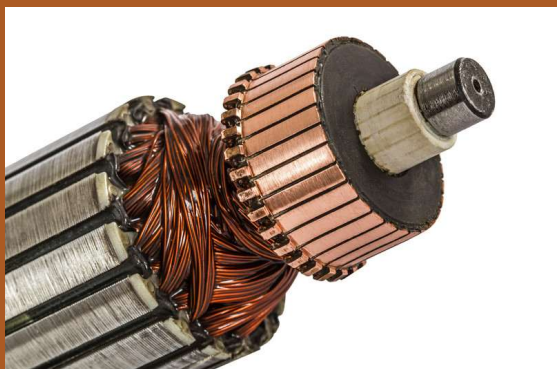


Worksheet 9

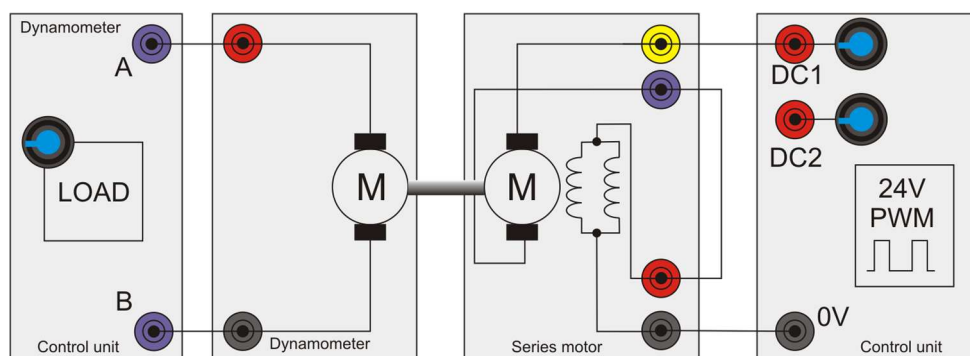
Series-wound motor characteristics

Modern electrical machines



A series-wound motor has the highest torque on start-up of any motor. However, its speed varies as the load varies. It is ideally suited to applications such as winches and cranes where the initially high torque is important.

The photograph shows a typical wound rotor with split ring commutator.



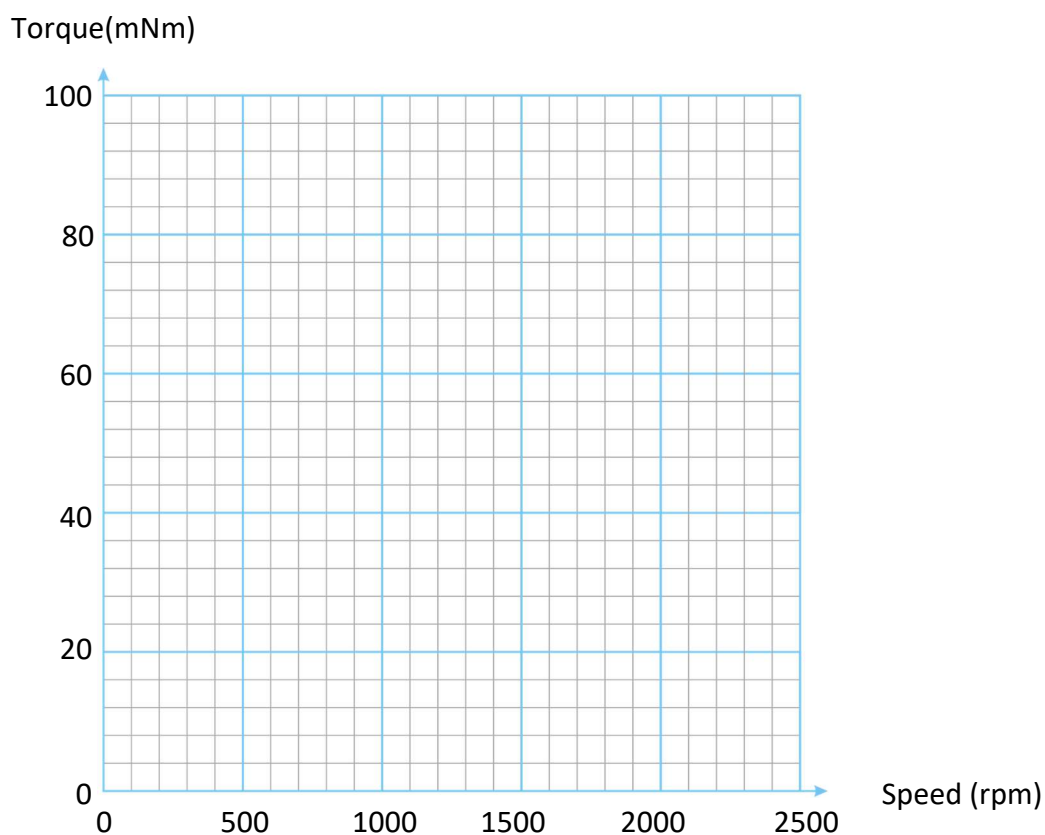
Over to you:

- 1) Set up the series motor as shown above.
- 2) On your PC, run the program 'Open Log_DC.bat'.
- 3) Run the program with the DC output set to around 35%.
- 4) Make sure the motor is putting pressure on the load cell rather than the balance - just reverse the direction of rotation if it is not.
- 5) Press 'Sweep' to take an automatic set of readings for a range of dynamometer loads.
- 6) Repeat steps 3, 4 and 5 for DC outputs of 40%, 45%, 50%, 55%.
- 7) When the sequence has finished load the results into Excel and use Excel to plot a graph of speed vs torque.
- 8) On the next page sketch graphs to show traces for all output voltages.

Worksheet 9

Series wound motor characteristics

Modern electrical machines



The effect of speed on torque for different values of DC output voltage.

Over to you:

- 1) With the same equipment set up, load the program 'Open Control_DC.bat'.
- 2) Route the Dynamometer output to the control unit Voltmeter. The software gives Dynamometer current. Now you can get Dynamometer voltage and current and calculate power in.
- 3) Run the program and set the DC1 output to around 30%
- 4) Complete the table to show speed, output power, dynamometer power and efficiency.

Power supply DC1	Speed rpm	Power out	Dyno current	Dyno Voltage	Efficiency
30%					
70%					